Bending an ice shelf with meltwater

A 2-D viscoelastic beam model: derivation and numerics

11 November 2026

This note derives, from scratch, a simple model of how a floating ice shelf bends when a meltwater lake fills (and drains) on its surface, and then shows how to solve it on a computer in a few lines. Every step comes with a picture. The code is in `flexure/viscoelastic_beam.py`. The figures come from `flexure/make_diagrams.py` and `flexure/run_examples.py`.

The whole model in two lines. The shelf is a thin Maxwell (spring-and-dashpot) beam floating on sea water. In its simplest form the deflection $w(x,t)$, positive downwards, obeys

$$\left(D\,\frac{\partial^4}{\partial x^4}+k\right)\frac{\partial w}{\partial t} \;=\; \frac{\partial q}{\partial t} \;+\; \frac{q-k\,w}{\tau},$$

where $q=\rho_w g\,d$ is the weight of a lake of depth d, $k=\rho_{sw}g$ is the stiffness of buoyancy, D is the flexural rigidity and τ is the Maxwell relaxation time. When a lake first appears, the ice responds **elastically**. Over a few days it relaxes towards **local isostasy**, $w=q/k$.

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The model actually solved goes one step further. Ice creeps without changing volume, so its viscous Poisson's ratio is $\frac{1}{2}$, while its elastic one is about 0.3. That splits the bending moment into two parts with relaxation times of 4.9 and 3.0 days (§5). The equation above is the special case where the two Poisson's ratios are equal. The split changes deflections by less than 1% and stresses by up to about 20% (in map view).

Version 2. This edition adds material from the working sessions on the derivation: the sign conventions and a free-end check (§3); the geometric origin of $\varepsilon = z\kappa$, the plane-strain modulus, and the naming of D (§4); a step-by-step derivation of the viscous factor 4η (§4); why the Maxwell rates add in series (§5); the assumptions behind the master equation (§6); a step-by-step derivation of the relaxation spectrum, a comparison of the Fourier and real-space routes, and a worked code block (§8); and the thin-plate caveat for narrow lakes (§8, §13). No equation in §1–§7 or §9–§14 has changed.

1 Set-up and assumptions

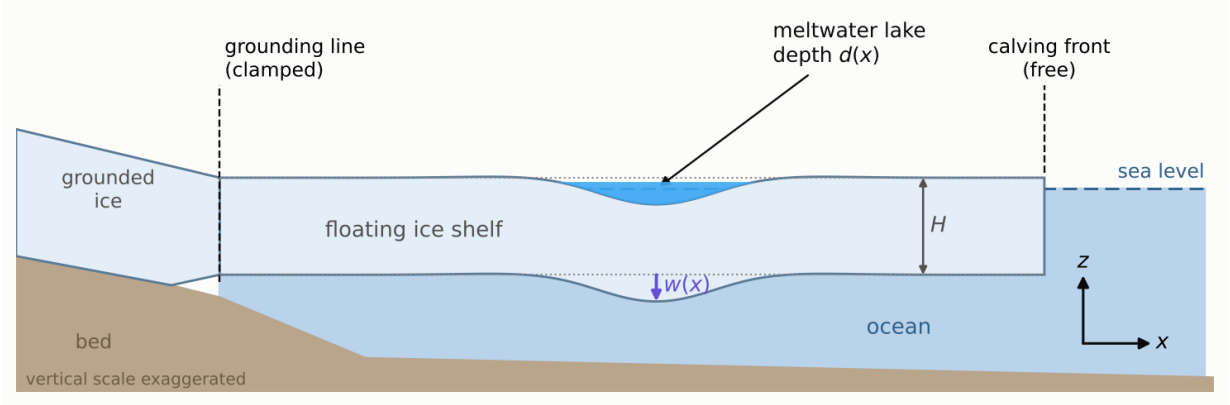


Figure 1: A vertical slice through the problem. A meltwater lake of depth $d(x)$ sits on a floating ice shelf of thickness H, which deflects downwards by $w(x)$. Vertical scale exaggerated.

We take a vertical slice through the shelf along a line x that runs from the grounding line towards the calving front. z points up. The shelf has thickness H, floats in sea water, and carries a lake of water depth $d(x,t)$. The lake's weight pushes the shelf down by $w(x,t)$, measured **positive downwards**. The change in surface elevation is therefore $-w$.

We assume the following:

assumption	why it is reasonable
2-D, plane strain. Nothing varies along the third direction y, so the lake is a long trough.	It keeps the maths one-dimensional. A circular lake gives similar numbers (see §13).
Thin beam. Plane cross-sections stay plane.	Lakes are 1 to 5 km across and $H \approx 100$ to 500 m.
Small deflections. $w \ll H$, small slopes.	w is a few metres.
Linear Maxwell rheology.	This is the simplest material that is elastic over hours and viscous over weeks.
Hydrostatic ocean. Sea water pushes back like a bed of springs.	Lakes fill and drain over hours to weeks, much slower than ocean waves.
Prescribed load $q = \rho_w g d(x,t)$.	We specify the water and do not route it (see §13).

assumption	why it is reasonable
No stretching or flow of the shelf.	Bending happens much faster than the shelf spreads.

Everything is linear in the **perturbation**. The background weight of the ice is already balanced by the background buoyancy, so only the extra load q and the extra deflection w appear.

2 Buoyancy is a bed of springs

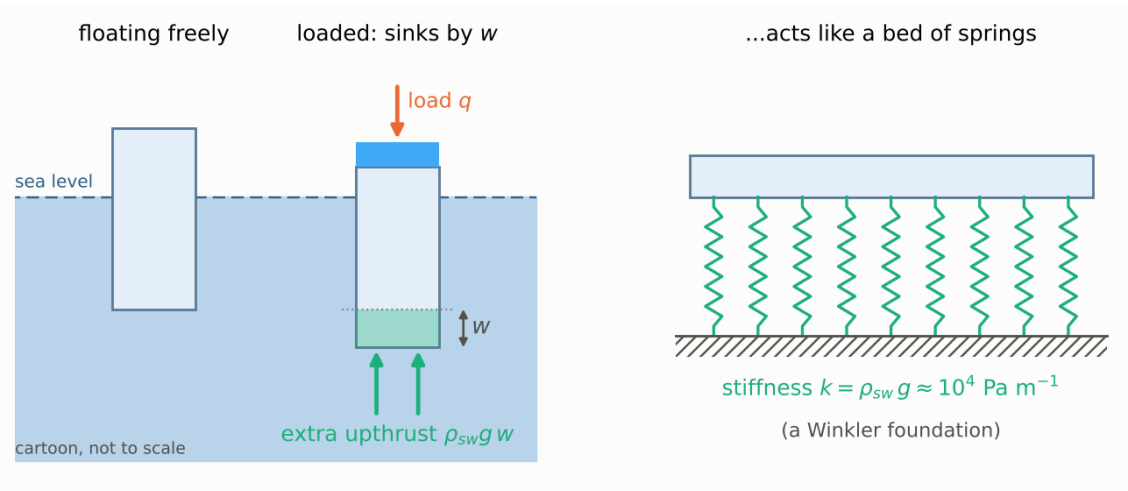


Figure 2: Pushing a floating column down by w displaces extra sea water and produces an extra upthrust $\rho_{sw} g\, w$. Buoyancy therefore acts like a bed of springs of stiffness $k = \rho_{sw} g$.

Push a floating column down by w and it displaces an extra w of sea water per unit area. Archimedes then gives an extra upward pressure

$$p_{\text{buoy}} = \rho_{sw}\, g\, w \;\equiv\; k\, w, \qquad k = \rho_{sw} g \approx 1.0 \times 10^4 \text{ Pa m}^{-1}.$$

This acts exactly like a bed of independent springs with stiffness k, which is called a *Winkler foundation*. The spring is the ocean, not the ice.

A useful by-product: if nothing else resisted, a lake of depth d would sink the ice by

$$w_{\text{iso}} = \frac{q}{k} = \frac{\rho_w}{\rho_{sw}}\, d \approx 0.97\, d.$$

3 Force balance on a slice

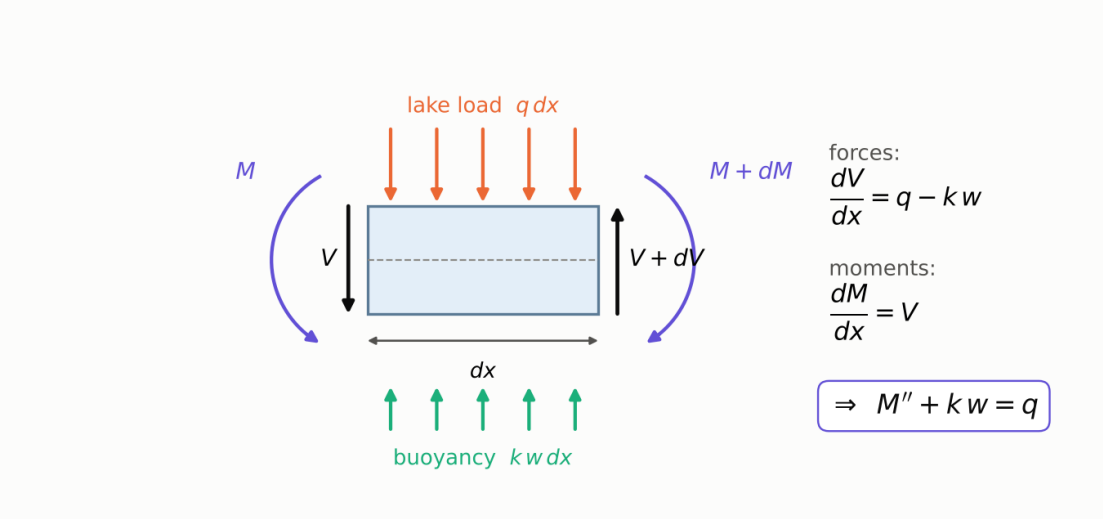


Figure 3: Forces and moments on a slice of the shelf of width dx.

Cut out a slice of width dx. Four things act on it:

- the lake load $q\,dx$, pushing down;
- buoyancy $k\,w\,dx$, pushing up;
- shear forces V and $V + dV$ on its two faces, which come from the neighbouring ice;
- bending moments M and $M + dM$ on its two faces.

Vertical forces must balance: $dV = (q - kw)\,dx$. **Moments** must balance (taken about the left face, ignoring tiny dx^2 terms): $dM = V\,dx$. Eliminating V gives the equilibrium equation

$$\boxed{\frac{\partial^2 M}{\partial x^2} + k\,w = q} \tag{1}$$

In words: wherever the lake is not held up by buoyancy, the ice must carry the difference through changes in its bending moment.

3.1 Sign conventions, and a free-end check

Throughout, w is positive downwards and z is positive upwards. The shear force V and moment M are defined so that

$$\frac{\partial M}{\partial x} = V, \qquad \frac{\partial V}{\partial x} = q - k\,w.$$

Only these two relations matter. Which face's shear is called "positive" is a choice of convention. The equations, not a picture of rotation, fix the sign. (Earlier working notes used sagging-positive curvature and moment, which flips the signs of M and V but not the physics.)

A check. Take a free end at $x = 0$, so $M(0) = V(0) = 0$, with no buoyancy and a downward load $q > 0$. Then

$$V(x) = qx, \qquad M(x) = \tfrac{1}{2} q x^2.$$

The piece of plate between 0 and x carries a downward load qx. The material to its right supplies the matching upward force, so the shear grows with distance from the free end. The two shear forces on an element form a couple of size $V\,dx$, and that couple is balanced by the change in moment, $M(x+dx) - M(x)$. The element does not rotate, so "clockwise" or "anticlockwise" is not what fixes the sign.

4 Bending: curvature, strain and moment

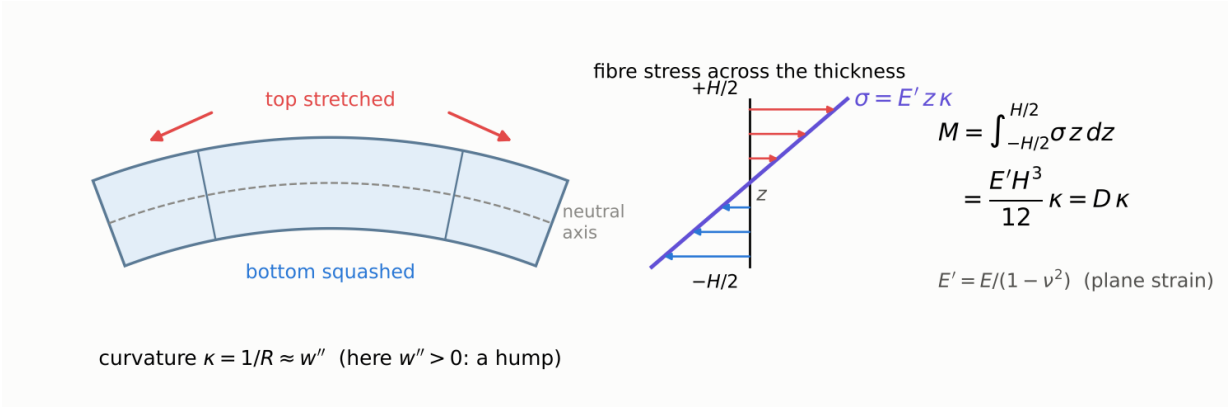


Figure 4: Cross-sections stay plane as the beam bends, so fibre strain and stress vary linearly with height z. Integrating σz across the thickness gives $M = D\kappa$.

When the beam bends, cross-sections stay straight but tilt. A fibre at height z above the mid-plane (the *neutral axis*) is stretched in proportion to its distance from it:

$$\varepsilon(z) = z\,\kappa, \qquad \kappa = \frac{\partial^2 w}{\partial x^2} \quad \text{(the curvature)}.$$

With w positive down, $\kappa > 0$ means a hump. The top is stretched (tension) and the bottom squashed. Under the middle of a lake the ice sags, so $\kappa < 0$ and the top is in compression.

The **bending moment** is the turning effect of all the fibre stresses $\sigma(z)$:

$$M = \int_{-H/2}^{H/2} \sigma(z)\, z\, dz.$$

4.0.1 Where $\varepsilon = z\kappa$ comes from

Here z is measured **upwards** from the mid-plane, which keeps the sign conventions of §4. Under a lake the ice sags, so $\kappa = \partial^2 w/\partial x^2 < 0$ and the top is in compression.

Take a short piece of plate that subtends a small angle $d\theta$ at the centre of its bending arc. Under sag, that centre lies above the plate.

- The mid-plane sits at radius R, so its length is $R\,d\theta$.
- A fibre at height z sits at radius $R - z$, so its length is $(R - z)\,d\theta$.

Strain is the change in length divided by the original length:

$$\varepsilon = \frac{(R - z)\,d\theta - R\,d\theta}{R\,d\theta} = -\frac{z}{R}.$$

With $|\kappa| = 1/R$ and $\kappa < 0$ for sag, this is $\varepsilon = z\kappa$. The $d\theta$ cancels, so the strain depends only on the distance from the mid-plane and on how tightly the plate is bent.

Two assumptions carry this result.

- **Plane sections stay plane and normal to the mid-plane** (the Kirchhoff assumption). This neglects shear through the thickness, which is fine when the plate is thin compared with the bending wavelength, $k_x H \ll 1$.
- **The mid-plane keeps its length.** There is no net in-plane force, so $\int \sigma \, dz = 0$. With a uniform modulus, the zero-strain surface then sits at the mid-plane. A stiffness that varies with depth (a cold, stiff top and a warm, soft base) moves it.

For small slopes the exact curvature $\kappa = w''/[1 + (w')^2]^{3/2}$ reduces to $\kappa \approx w''$.

4.0.2 The plane-strain modulus $E' = E/(1-\nu^2)$

For an isotropic elastic solid, each normal strain depends on all three normal stresses:

$$\varepsilon_{xx} = \tfrac{1}{E}\big[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})\big], \quad \varepsilon_{yy} = \tfrac{1}{E}\big[\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})\big], \quad \varepsilon_{zz} = \tfrac{1}{E}\big[\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})\big].$$

Two conditions simplify these.

- $\sigma_{zz} \approx 0$**.** The top and base are nearly free. The real surface loads are of order $\rho g H$, far smaller than the bending stresses, which exceed them by roughly $(L/H)^2$, where L is the bending wavelength.
- $\varepsilon_{yy} = 0$**.** The plate is infinite and uniform in y, so neighbouring slices must deform identically. A fibre stretched in x would like to contract in y, but its neighbours want to do the same, so that contraction is prevented.

Setting $\varepsilon_{yy} = 0$ and $\sigma_{zz} = 0$ in the yy equation gives $\sigma_{yy} = \nu\,\sigma_{xx}$. Substituting into the xx equation gives $\varepsilon_{xx} = \frac{1-\nu^2}{E}\sigma_{xx}$, that is,

$$\sigma_{xx} = E'\,\varepsilon_{xx}, \qquad E' = \frac{E}{1-\nu^2}.$$

The induced σ_{yy} acts back on x through ν a second time, which is why ν appears squared.

The size of the effect depends on the geometry. A narrow beam has free sides, so $\sigma_{yy} = 0$ and its modulus is E. A wide plate has locked sides and modulus E'. For ice with $\nu \approx 0.3$ the factor is 1.10, a 10% correction to D and under 3% to $\ell \propto D^{1/4}$. For an incompressible material ($\nu = \frac{1}{2}$) it is $4/3$. This E' is the modulus that enters D below.

4.0.3 What D is called

D is the **flexural rigidity**, also called the *bending stiffness* or *plate rigidity*. It is the bending moment per unit width needed to produce unit curvature, $M = D\kappa$, and its units are N m, equivalently Pa m^3. It is the plate version of EI for a narrow beam.

In geophysics people often quote the **effective elastic thickness** T_e instead. That is the value of H that reproduces an observed D through $D = EH^3/12(1-\nu^2)$. The viscous counterpart, $\Lambda = \eta H^3/3$, is called the **viscous flexural rigidity**; some notes write it D_v.

For an **elastic** beam in plane strain, $\sigma = E'\varepsilon$ with $E' = E/(1-\nu^2)$. So

$$M = E'\kappa \int_{-H/2}^{H/2} z^2\, dz = \frac{E'H^3}{12}\,\kappa \equiv D\,\kappa, \qquad D = \frac{EH^3}{12(1-\nu^2)}.$$

D is the **flexural rigidity**. It scales as H^3, so a shelf twice as thick is eight times as stiff.

For a **viscous** beam, stress depends on strain *rate*. For an incompressible Newtonian fluid of viscosity η in plane strain with $\sigma_{zz} \approx 0$, this is $\sigma = 4\eta\,\dot{\varepsilon}$. The same integral gives

$$M = \frac{4\eta H^3}{12}\,\dot{\kappa} \equiv \Lambda\,\dot{\kappa}, \qquad \Lambda = \frac{\eta H^3}{3}.$$

Λ is a "viscous flexural rigidity".

4.1 Deriving the viscous factor 4η

For an incompressible Newtonian fluid the stress is

$$\sigma_{ij} = -p\,\delta_{ij} + 2\eta\,\dot{\varepsilon}_{ij},$$

where p is a pressure. In an incompressible fluid p is not set by the deformation. It takes whatever value the constraints require. Three constraints apply in plane strain.

1. **Incompressibility:** $\dot{\varepsilon}_{xx} + \dot{\varepsilon}_{yy} + \dot{\varepsilon}_{zz} = 0$.
2. **Plane strain:** $\dot{\varepsilon}_{yy} = 0$.
3. **Free surfaces:** $\sigma_{zz} = 0$.

From (1) and (2), $\dot{\varepsilon}_{zz} = -\dot{\varepsilon}_{xx}$. A fibre stretching in x cannot narrow in y, so all the compensating contraction happens in z: the fibre thins vertically at the rate it lengthens.

From (3), $\sigma_{zz} = -p + 2\eta\dot{\varepsilon}_{zz} = 0$, so $p = 2\eta\dot{\varepsilon}_{zz} = -2\eta\dot{\varepsilon}_{xx}$. Then

$$\sigma_{xx} = -p + 2\eta\,\dot{\varepsilon}_{xx} = 2\eta\dot{\varepsilon}_{xx} + 2\eta\dot{\varepsilon}_{xx} = 4\eta\,\dot{\varepsilon}_{xx}.$$

This is the viscous counterpart of the elastic modulus E': the same plane-strain stiffening, applied to a fluid. The thickness integral then gives $M = \int \sigma z\, dz = \frac{4\eta H^3}{12}\dot{\kappa} = \Lambda\dot{\kappa}$, with $\Lambda = \eta H^3/3$.

Check. A rod of incompressible fluid pulled at strain rate $\dot{\varepsilon}$ carries stress $3\eta\dot{\varepsilon}$, so the uniaxial viscous modulus is 3η. Treat it like an elastic solid with $E \to 3\eta$ and $\nu \to \frac{1}{2}$ in $E' = E/(1-\nu^2)$:

$$\frac{3\eta}{1-\frac{1}{4}} = 4\eta.$$

This agrees with the plane-strain factor used above.

Caveat. Uniform η is assumed. For real ice η depends strongly on temperature, so $\Lambda = \int 4\eta(z)\, z^2\, dz$ about the neutral surface. The neutral surface moves towards the stiff, cold top, and the effective rigidity is dominated by the coldest layers.

5 Ice is a Maxwell material

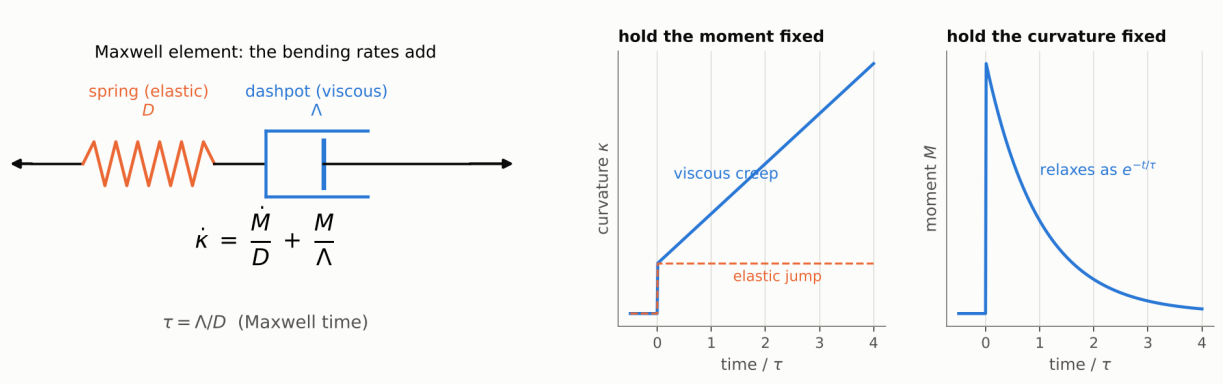


Figure 5: A Maxwell element (a spring and a dashpot in series) and its two classic responses: creep under a fixed moment, and relaxation at fixed curvature.

Over minutes to hours, ice behaves like a spring. Over days to weeks it creeps like a very stiff fluid. The simplest model that does both is a **Maxwell element**, a spring and a dashpot in series. Each fibre obeys $\dot{\varepsilon} = \dot{\sigma}/E' + \sigma/4\eta$. Multiply by z and integrate across the thickness, exactly as above, and the *bending* rates simply add:

$$\boxed{\dot{\kappa} = \frac{\dot{M}}{D} + \frac{M}{\Lambda}} \qquad \Longleftrightarrow \qquad \dot{M} = D\,\dot{\kappa} - \frac{M}{\tau}, \tag{2}$$

with the **Maxwell time**

$$\tau = \frac{\Lambda}{D} = \frac{4\eta(1-\nu^2)}{E}.$$

For $\nu \to \frac{1}{2}$ this becomes the familiar η/G, where G is the shear modulus. With $E = 1$ GPa and $\eta = 10^{14}$ Pa s, $\tau \approx 4.2$ days. The right-hand plots show the two classic experiments:

- **Hold M fixed.** κ jumps elastically and then creeps linearly.
- **Hold κ fixed.** M jumps and then relaxes as $e^{-t/\tau}$.

A useful mental picture is an **elastic beam whose stress-free shape slowly drifts towards whatever shape it is being held in**.

5.1 Why the two rates add (series, not parallel)

The Maxwell element is a spring and a dashpot joined end to end, so both carry the same stress and their extensions add. For one fibre,

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E'} + \frac{\sigma}{4\eta},$$

with the spring giving $\dot{\varepsilon}_e = \dot{\sigma}/E'$ and the dashpot giving $\dot{\varepsilon}_v = \sigma/4\eta$.

This arrangement matches how ice behaves.

- **Sudden load.** The dashpot cannot move instantly, so the spring takes all the load and the response is immediately elastic.

- **Sustained load.** The dashpot keeps extending for as long as stress remains, so the ice flows indefinitely.

A parallel arrangement (Kelvin–Voigt) adds the stresses instead. It creeps towards a fixed elastic strain and then stops, which is not what ice does.

The bending kinematics apply to the total strain, because plane sections stay plane: $\varepsilon = z\kappa$, so $\dot{\varepsilon} = z\dot{\kappa}$. Substituting this into the Maxwell law and integrating over the thickness, with the weighting z, gives $\dot{\kappa} = \dot{M}/D + M/\Lambda$. That is equation (2). The curvature rates add because the total curvature rate is the sum of an elastic rate, $\dot{M}/D$, and a viscous rate, M/Λ.

What this means.

- The moment sets the *rate* of bending, not the curvature itself. A constant moment makes the curvature grow linearly in time, with no equilibrium shape. That is why the viscous plate keeps sagging until buoyancy takes over the whole load.
- The units differ: Λ is in Pa s m^3 and D in Pa m^3. So Λ/D is a time, the Maxwell time τ.
- Uniform η is assumed, as in the viscous factor above.

5.2 Two Poisson's ratios, two relaxation times

So far the spring and the dashpot share the same sideways behaviour. They should not. Elastically, ice has a Poisson's ratio $\nu \approx 0.3$. Creep, however, conserves volume, so the viscous part behaves as if its Poisson's ratio were $\nu_v = \frac{1}{2}$ (as in MacAyeal et al., 2021, and Bézu et al., 2025). A single spring–dashpot pair cannot have two Poisson's ratios.

The fix comes from the plate's full moment tensor. Split it into an **isotropic** part (equal bending in every direction) and a **deviatoric** part (more bending one way than the other).

- The elastic stiffnesses of the two parts are $D(1+\nu)$ and $D(1-\nu)$.
- The viscous ones are $\Lambda(1+\nu_v)$ and $\Lambda(1-\nu_v)$.
- Each part is its own Maxwell element.

A beam in plane strain ($\kappa_{yy} = 0$) bends both parts, with weights $(1 \pm \nu)/2$. Its moment is therefore the sum of **two Maxwell elements in parallel**:

$$M = M_1 + M_2, \qquad \dot{M}_i = D_i\,\dot{\kappa} - \frac{M_i}{\tau_i}, \qquad D_{1,2} = D\,\frac{1 \pm \nu}{2}, \tag{2′}$$

$$\tau_1 = \tau\,\frac{1+\nu_v}{1+\nu} = \frac{4\eta(1+\nu_v)(1-\nu)}{E}, \qquad \tau_2 = \tau\,\frac{1-\nu_v}{1-\nu} = \frac{4\eta(1-\nu_v)(1+\nu)}{E}.$$

For $\nu_v = \frac{1}{2}$ these are $\tau_1 = 6\eta(1-\nu)/E \approx 4.9$ days and $\tau_2 = 2\eta(1+\nu)/E \approx 3.0$ days. Four checks:

- **Elastic limit:** $M = (D_1 + D_2)\kappa = D\kappa$, as before.
- **Viscous limit:** $M = (D_1\tau_1 + D_2\tau_2)\dot{\kappa} = \Lambda\dot{\kappa}$, the same viscous rigidity as before.
- **Mean time:** the rigidity-weighted mean $(D_1\tau_1 + D_2\tau_2)/D$ is still $\tau = \Lambda/D \approx 4.2$ days.
- **Equal Poisson's ratios:** if $\nu_v = \nu$ the two times are equal and (2′) collapses back to the single element (2).

In map view the same split gives the viscous plate law $M = \Lambda\big[(1-\nu_v)\dot{\boldsymbol{\kappa}} + \nu_v\,\mathrm{tr}(\dot{\boldsymbol{\kappa}})\,I\big]$. For $\nu_v = \frac{1}{2}$ this is exactly the viscous bending law used by MacAyeal et al. (2021) and Bézu et al. (2025).

5.3 Relation to MacAyeal, Sergienko & Banwell (2015)

MacAyeal et al. (2015) wrote down the thin-plate viscoelastic model of ice-shelf flexure under meltwater lakes, motivated by the 2002 break-up of the Larsen B Ice Shelf. Physically, the model in this document is the same one.

Same framework. One bending-moment field drives an elastic spring and a viscous dashpot in series, and the plate floats on sea water. Their "buoyancy bucket" is our bed of springs, $k = \rho_{sw} g$.

Same plate laws. They split the curvature $\mathbf{H} = (\kappa_{xx}, \kappa_{yy}, \kappa_{xy})$ into elastic and viscous parts, driven by the same moment $\mathbf{M} = (M_{xx}, M_{yy}, M_{xy})$:

$$\dot{\mathbf{H}} = \mathbf{D}^{-1}\dot{\mathbf{M}} + \mathbf{V}^{-1}\mathbf{M}, \qquad \mathbf{D} = D\begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & 1-\nu \end{pmatrix}, \qquad \mathbf{V} = \Lambda\begin{pmatrix} 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}.$$

This is their eqs 17–19. It is written here in our notation and sign convention: their displacement is positive upwards, their viscosity is our η, and $\Lambda = \eta H^3/3$. $\mathbf{D}$ is our elastic plate law, and $\mathbf{V}$ is our viscous plate law with $\nu_v = \frac{1}{2}$.

Our two relaxation times are their equation, diagonalised. $\mathbf{D}$ and $\mathbf{V}$ share three eigenvectors: isotropic bending $(1, 1, 0)$, deviatoric bending $(1, -1, 0)$ and twist $(0, 0, 1)$. Their eigenvalues are $D(1+\nu)$, $D(1-\nu)$, $D(1-\nu)$ and $\Lambda(1+\nu_v)$, $\Lambda(1-\nu_v)$, $\Lambda(1-\nu_v)$. In that basis the matrix equation falls apart into independent Maxwell elements, $\dot{M} = d\,\dot{\kappa} - M/\tau$, with

$$\tau_1 = \frac{\Lambda(1+\nu_v)}{D(1+\nu)} \text{ (isotropic)}, \qquad \tau_2 = \frac{\Lambda(1-\nu_v)}{D(1-\nu)} \text{ (deviatoric and twist)},$$

which are the times of (2′). Any model with one moment field and two Poisson's ratios therefore has two relaxation times; (2′) just makes them explicit.

Their experiment, rerun. They filled a circular lake, 500 m in radius and 2 m deep, on a 200 m thick shelf ($E = 10$ GPa, $\nu = 0.3$) over 60 days, drained it in 6 hours, and followed the shelf for a year. Their ice obeys Glen's law with $n = 3$. The table repeats this with `SpectralPlate` (a 25.6 km periodic box at 50 m, exact in time) for three constant viscosities. It gives the depression at the lake centre; their values are read from their Figs 4 and 5.

	day 60 (full)	just after drainage	day 365
MacAyeal et al. (2015), Glen's law $n = 3$	58 cm	36 cm	19 cm
this model, $\eta = 5 \times 10^{13}$ Pa s	150 cm	126 cm	0.4 cm
this model, $\eta = 3 \times 10^{14}$ Pa s	83 cm	63 cm	4.0 cm
this model, $\eta = 2 \times 10^{15}$ Pa s	41 cm	21 cm	3.5 cm

The elastic jump when the water leaves agrees. Ours is the purely elastic response to the lake, 19.8 cm, plus up to 4 cm of creep during the 6 hours of drainage when η is low; theirs is about 22 cm. The recovery does not agree. With a constant viscosity the centre recovers to within a few centimetres over the following year, and no single value of η matches both their filling and their recovery. Under Glen's law the effective viscosity rises as the stresses fall, so their recovery

stalls with 19 cm still to go. Their suggestion that repeated fill–drain cycles deepen lake basins year after year therefore rests on the nonlinear rheology.

Differences.

- *Rheology.* They offer Glen's law, through an effective viscosity built from the invariants of **M** (their eqs 39–40). Here η is constant (see §13).
- *Parameters.* They take $E = 10$ GPa; here $E = 1$ GPa, from tidal flexure. For the same η that makes ℓ about 1.8 times shorter and τ 10 times longer.
- *Unknowns and numerics.* They solve for **M**, the viscous curvature and the deflection (seven unknowns) with COMSOL finite elements, and demonstrate one axisymmetric lake with a simply supported edge at $r = 10$ km. Here the 1-D beam uses the scheme of §11. The 2-D plates carry two scalar moment potentials instead of the tensor (see `flexure/viscoelastic_plate.py`), solved by finite differences with grounded ice held fixed, or by FFTs exactly in time.
- *Thickness.* Their tensor form takes spatially varying thickness and viscosity as it stands. The moment potentials need a uniform rigidity in 2-D (see §13).
- *Scope.* They study one idealised lake for one year. Here lakes are routed by fill-spill-merge on real REMA surfaces, with grounding lines, and the routing can be coupled to the flexure.

6 The master equation

Now combine the force balance (1) with the material law. Take $\partial^2/\partial x^2$ of each Maxwell law in (2′), add them, and use (1) differentiated in time, $\dot{M}'' = \dot{q} - k\dot{w}$:

$$\dot{q} - k\,\dot{w} \;=\; D\,\frac{\partial^4 \dot{w}}{\partial x^4} \;-\; \frac{\partial^2}{\partial x^2}\Big(\frac{M_1}{\tau_1} + \frac{M_2}{\tau_2}\Big).$$

Rearranged:

$$\boxed{\Big(D\,\frac{\partial^4}{\partial x^4} + k\Big)\frac{\partial w}{\partial t} = \frac{\partial q}{\partial t} + \frac{\partial^2}{\partial x^2}\Big(\frac{M_1}{\tau_1} + \frac{M_2}{\tau_2}\Big)} \tag{3}$$

When the two times are equal ($\nu_v = \nu$), the last term is M''/τ. By (1) that is $(q - kw)/\tau$, and (3) becomes the **single-element master equation**

$$\Big(D\,\frac{\partial^4}{\partial x^4} + k\Big)\frac{\partial w}{\partial t} = \frac{\partial q}{\partial t} + \frac{q - k\,w}{\tau}. \tag{3′}$$

Read (3′) left to right:

- The operator in brackets is the **elastic** beam-on-springs operator.
- $\partial q/\partial t$ is the **instantaneous elastic** response to any change in load.
- $(q - kw)/\tau$ is the **creep** term. Wherever buoyancy is not yet carrying the lake ($kw \neq q$), the ice creeps at a rate set by τ, and it stops only when $kw = q$.

The full equation (3) says the same thing, except that the two parts of the bending moment creep away at slightly different rates.

6.1 Assumptions behind the master equation

- **Thin plate.** The plate is much thinner than the load width, $k_x H \ll 1$. This fails for narrow lakes on thick shelves (see §8.5).
- **Small slopes.** The geometry is linear, with $\kappa \approx w''$.
- **No in-plane force.** An axial force would add a term $N\,\partial^2 w/\partial x^2$ to the equation.
- **Plane strain.** The geometry is a 2-D strip, infinite in y.
- **Quasi-static.** Inertia is neglected.
- **Uniform properties.** H, E and η are uniform. Temperature-dependent viscosity makes the viscous rigidity depth-weighted rather than $\eta H^3/3$.

Read these with the master equation (3′) in mind. Each one is an assumption about the plate or the load. None of them is a claim about what happens after the load changes.

7 Two limits: elastic and isostatic

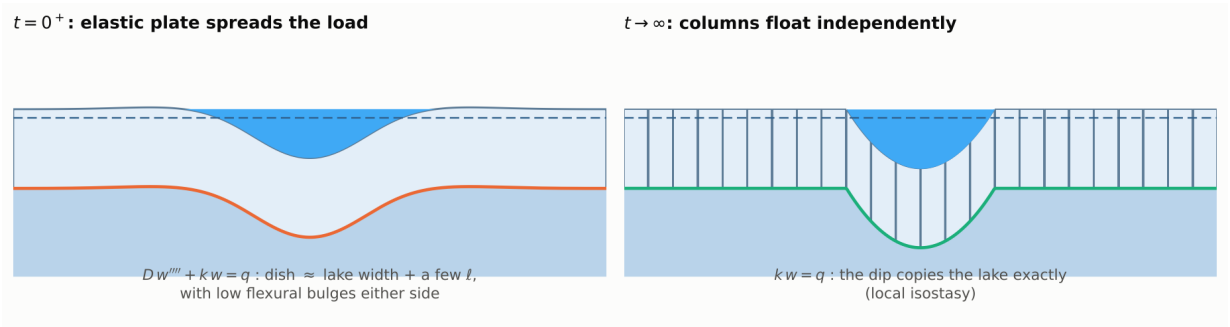


Figure 6: The two limits of the model: an elastic response at $t = 0^+$ and local isostasy as $t \to \infty$.

At $t = 0^+$ the response is elastic. Integrate (3) across the instant the load appears. The creep term is finite, so it contributes nothing in zero time, and we are left with

$$D\,\frac{\partial^4 w}{\partial x^4} + k\,w = q.$$

This is the classic "beam on elastic foundation" (Hetényi, 1946). Its natural length scale is the **flexural length**

$$\ell = \left(\frac{D}{k}\right)^{1/4} \approx 610\ \text{m} \quad (H = 250\ \text{m},\ E = 1\ \text{GPa}).$$

For a line load P the solution is $w = \frac{P\beta}{2k}\,e^{-\beta|x|}\big(\cos\beta|x| + \sin\beta|x|\big)$, with $\beta = 1/(\sqrt{2}\,\ell)$.

- The dip is spread over a few flexural lengths.
- It crosses zero at $|x| = 3\pi\ell/(2\sqrt{2}) \approx 3.3\ell$.
- A low **flexural bulge** peaks at $|x| = \sqrt{2}\,\pi\,\ell \approx 4.4\ell$, with height $e^{-\pi} \approx 4\%$ of the central dip.

Our 3 km-wide, 4 m-deep lake sinks the centre by 3.2 m at first, instead of 3.9 m.

As $t \to \infty$ the response is isostatic. Once nothing changes any more ($\dot{w} = 0$), each Maxwell law makes its M_i decay to zero, so (1) gives $q = kw$. The ice "forgets" its stiffness and floats as independent columns. The dip copies the lake exactly, and the bending moment relaxes to zero.

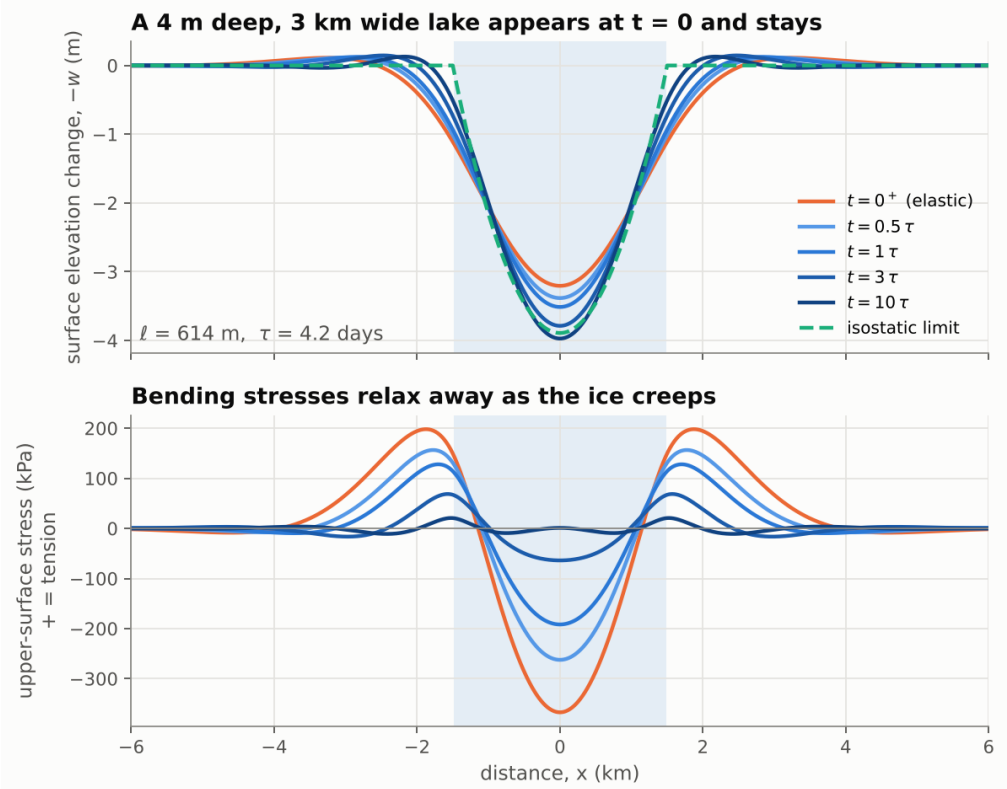


Figure 7: A lake appears at $t = 0$ and is left there. The orange curve is the elastic response. The blue curves (light to dark) show the ice creeping towards the dashed isostatic shape. The bending stresses (bottom panel) decay as it does.

8 Which shapes relax first? The relaxation spectrum

Because the equations are linear with constant coefficients, each sinusoid in x evolves on its own. A load with any shape can be split into sinusoids, each relaxes at its own rate, and the total response is their sum. This section derives the rate for one sinusoid, shows what it means, and compares this route with the real-space solution.

8.1 8.1 One wavelength, step by step

Put $w = \hat{w}(t)\cos k_x x$ and $q = \hat{q}\cos k_x x$, with $\hat{q}$ switched on at $t = 0$ and held constant. Since $\partial_x^4 \cos k_x x = k_x^4 \cos k_x x$, and $\partial_t q = 0$ for $t > 0$, the single-element master equation (3′) becomes an ordinary differential equation:

$$\left(Dk_x^4 + k\right)\frac{d\hat{w}}{dt} = \frac{\hat{q} - k\,\hat{w}}{\tau}.$$

Step 1: the gap. Define the shortfall from isostasy,

$$y(t) \equiv \hat{w}(t) - \frac{\hat{q}}{k}.$$

It is negative while the plate has not yet sunk far enough.

Step 2: rewrite. $\hat{q}$ is constant for $t > 0$, so $\hat{q}/k$ is constant too, and $dy/dt = d\hat{w}/dt$.

Step 3: substitute. Since $\hat{q} - k\hat{w} = -k\,y$,

$$\left(Dk_x^4 + k\right)\frac{dy}{dt} = -\frac{k}{\tau}\,y \quad \Longrightarrow \quad \frac{dy}{dt} = -\frac{y}{\tau_\lambda}, \qquad \tau_\lambda = \tau\,\frac{Dk_x^4 + k}{k} = \tau\left[1 + (k_x\ell)^4\right],$$

using $\ell^4 = D/k$. The gap closes at a rate proportional to its own size.

Step 4: solve. Separating variables and integrating gives $y(t) = y_0\, e^{-t/\tau_\lambda}$, where y_0 is the gap at the moment creep starts.

Step 5: the starting gap. Creep starts just after the elastic jump, when $\hat{w}(0^+) = \hat{q}/(Dk_x^4 + k)$. So

$$y_0 = \frac{\hat{q}}{Dk_x^4 + k} - \frac{\hat{q}}{k}.$$

This is negative: the elastic deflection falls short of isostasy because bending carries part of the load.

Step 6: back to deflection.

$$\hat{w}(t) = \underbrace{\frac{\hat{q}}{k}}_{\text{isostatic}} + \Big(\underbrace{\frac{\hat{q}}{Dk_x^4 + k}}_{\text{elastic}} - \frac{\hat{q}}{k}\Big)\, e^{-t/\tau_\lambda}.$$

Step 7: checks. At $t = 0^+$ the exponential is 1, and the result is the elastic value. As $t \to \infty$ the exponential vanishes, and the result is isostatic. After one τ_λ the shortfall has fallen to $1/e \approx 37\%$ of its starting size. In the right-hand panel of the relaxation-spectrum figure, each curve starts at its elastic value and closes that fraction of the gap per τ_λ.

What it means.

- **Broad features** ($\lambda \gg 2\pi\ell$) are barely supported by bending in the first place. They are already nearly isostatic, and what is left relaxes in about one Maxwell time.
- **Narrow features** ($\lambda \lesssim 2\pi\ell$) are held up mainly by the ice's stiffness. They relax extremely slowly, as $(2\pi\ell/\lambda)^4$. On a 250 m-thick shelf, a 2 km-wavelength wiggle takes about two months to relax, and a 1 km one takes more than two years (thin-plate values; see §8.5).

8.2 8.2 Two Poisson's ratios

The elastic and viscous Poisson's ratios differ, so one wavelength is not a single Maxwell element. It is two coupled elements, one for each Poisson's ratio, and each wavelength therefore relaxes as the sum of two exponentials. Their exact form is a 2×2 matrix exponential per mode (`relaxation_times` and `modal_step_response` in the code). The left panel of the figure shows both times.

- The **slow mode** stays within 15% of τ_λ, and within 1% for narrow features. It carries most of the creep.
- The **fast mode** has a time of 3.0–3.5 days. It carries up to about a third of the (already small) relaxation of broad features, and almost none of the relaxation of narrow ones.

With two Poisson's ratios there is no single τ, so the figure normalises by a mean Maxwell time. That is why the deflection hardly changes. The right panel's dotted lines are the single-element model.

This also tells us that the problem is **not stiff** numerically. The fastest decay rate of any mode is $1/\tau_2$ (about 3 days), so a time step of a fraction of a day is enough for every wavelength.

8.3 8.3 Code for one mode

The response of one mode is a single function. This is the form used for the plots in this section.

```
import numpy as np

def w_hat(t, kx, q_hat, k, D, tau):
    """Amplitude of the cos(kx x) mode after a step load q_hat switched on at t = 0."""
    ell = (D / k) ** 0.25
    w_iso = q_hat / k                       # isostatic, t -> infinity
    w_el = q_hat / (D * kx**4 + k)          # elastic jump, t = 0+
    tau_l = tau * (1 + (kx * ell) ** 4)     # relaxation time of this mode
    return w_iso + (w_el - w_iso) * np.exp(-t / tau_l), tau_l
```

Pass an array of wavelengths as `kx` to get the relaxation spectrum. The function's second output, `tau_l`, is the same τ_λ as above. Its validity depends on $k_x H$; see §8.5.

8.4 8.4 Two routes to the same elastic response

The elastic response to a load can be found in two ways. The Fourier route is an algebraic shortcut. The real-space route gives the shape of the deflection around one particular load. Both give the same physics.

The Fourier route: algebra rather than calculus. A cosine passes through ∂_x^4 unchanged apart from a factor k_x^4. Substituting $w = \hat{w}\cos k_x x$ into $Dw'''' + kw = q$ turns the differential equation into an algebraic one:

$$\left(Dk_x^4 + k\right)\hat{w} = \hat{q} \quad \Longrightarrow \quad \hat{w} = \frac{\hat{q}}{Dk_x^4 + k} = \frac{\hat{q}/k}{1 + (k_x\ell)^4}.$$

No boundary conditions, symmetry arguments or force balance need to be applied. They are all built into the choice of a sinusoid on an infinite plate, which is bounded, has the right symmetry, and needs no patching. The result is a **transfer function**, the plate's response to each wavelength. It is a low-pass filter:

- **Long wavelengths** pass through at the isostatic value, $\hat{q}/k$.
- **Short wavelengths** are suppressed as $(k_x\ell)^{-4}$, because bending holds them up.

The real-space route: an explicit shape. Here the work is:

1. find the roots of $Dm^4 + k = 0$;
2. assemble the four decaying and oscillating solutions;
3. discard the growing ones;
4. apply symmetry, $w'(0) = 0$;
5. apply the force condition, $\int k\,w\,dx = P$.

This takes more work, but what comes out is a **shape**. For a line load P,

$$w = \frac{P\beta}{2k}\,e^{-\beta|x|}\left(\cos\beta|x| + \sin\beta|x|\right), \qquad \beta = \frac{1}{\sqrt{2}\,\ell}.$$

Step 4 fixes the constant. For $x > 0$ write $w = e^{-\beta x}(A\cos\beta x + B\sin\beta x)$. Near $x = 0$ the A term falls with slope $-\beta$, and the B term rises with slope $+\beta$, so $w'(0) = \beta(B - A)$. Symmetry requires $w'(0) = 0$, which gives $B = A$. The B term does not change the height at the load, but

it tilts the plate, and $B = A$ makes the upward tilt cancel the downward one so the base is flat at the load.

From the shape you can read off the dip, the zero crossing at $\beta x = 3\pi/4$ (that is, $|x| \approx 3.3\,\ell$), and the forebulge at $\beta x = \pi$ (that is, $|x| \approx 4.4\,\ell$). You also get closed forms for a lake of width W. Its centre deflection, in units of the isostatic deflection, is $1 - e^{-\beta W/2}\cos(\beta W/2)$, and at the shoreline the deflection is half the isostatic value. None of these is visible in the transfer function.

How they connect. The Green's function is the inverse Fourier transform of the transfer function. A point load contains every wavelength equally, so $\hat{q} = P$ for all k_x, and

$$w(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{P}{Dk_x^4 + k}\, e^{ik_x x}\, dk_x.$$

Doing this integral by contour integration moves the work into finding the poles, where $Dk_x^4 + k = 0$. With $m = ik_x$ these are the same complex roots $\beta(\pm 1 \pm i)$ as in real space. Closing the contour in the upper half-plane picks out the two decaying roots. This is the Fourier counterpart of discarding the growing exponentials. Both routes do the same work, and the Fourier route simply defers it to the inversion step.

Each real-space condition has a Fourier counterpart:

Real-space step	Fourier counterpart
Drop growing exponentials	Bounded sinusoids; contour closed on the correct side
Symmetry, $w'(0) = 0$	Cosines are already symmetric
$\int k\, w\, dx = P$	The $k_x = 0$ mode: $\hat{w}(0) = \hat{q}(0)/k$
Forebulge	Ringing from the filter's sharp roll-off at $k_x\ell \approx 1$

The force condition says something general. The zero-wavenumber mode is always isostatic, so **the total displaced volume is always load$/k$, whatever the stiffness**. Bending redistributes the deflection but cannot change its total. That is also why the forebulge exists: in the point-load solution the dip is deeper and wider than the load alone would need, and the bulge pays back the difference.

When each is the right tool.

- **Fourier** is better when time-dependence matters. Each mode becomes its own ODE, which gives the relaxation spectrum τ_λ almost for free, as in §8.1. It is also the natural way to see which scales dominate a load.
- **Real space** is better when you want the shape of one load: the depth of the dip, the zero crossing, the forebulge, or closed forms for a particular lake.
- **Boundaries.** The Fourier route assumes an infinite plate. A grounding line or calving front breaks that assumption, so those ends are handled with real-space conditions (§10).

8.5 8.5 Applying the spectrum to lakes, and thin-plate validity

λ is not lake width. A lake of width W corresponds roughly to $\lambda \approx 2W$. So a "1 km wiggle" corresponds to a lake about 500 m across.

The thin-plate condition. Everything in this section assumes $k_x H \ll 1$. At $\lambda = 1$ km on 250 m ice, $k_x H \approx 1.6$, which is already marginal. As $k_x H$ grows, the beam increasingly overestimates τ_λ. The λ^{-4} steepening is a thin-plate artefact. The exact layer solution in the code turns over to half-space scaling, $\tau_\lambda \propto \eta\lambda$, instead. So for narrow lakes on thick shelves the values in §8.1 are overestimates, and the thin-plate form should be checked against the full solution.

9 Stresses, and why draining is dangerous

The bending stress is largest at the upper surface, where crevasses open:

$$\sigma_{\text{top}} = \frac{6M}{H^2} \qquad (\text{positive} = \text{tension}).$$

For the 4 m-deep, 3 km-wide lake:

- The elastic response puts the surface under the lake centre into compression of about -370 kPa.
- A ring of **tension, about $+200$ kPa**, sits just outside the lake shore.
- These stresses are of the same order as those usually needed to open surface crevasses, roughly 100 kPa or more (e.g. Vaughan, 1993).

This is the "ring fracture" mechanism proposed for the 2002 Larsen B break-up (Banwell et al., 2013).

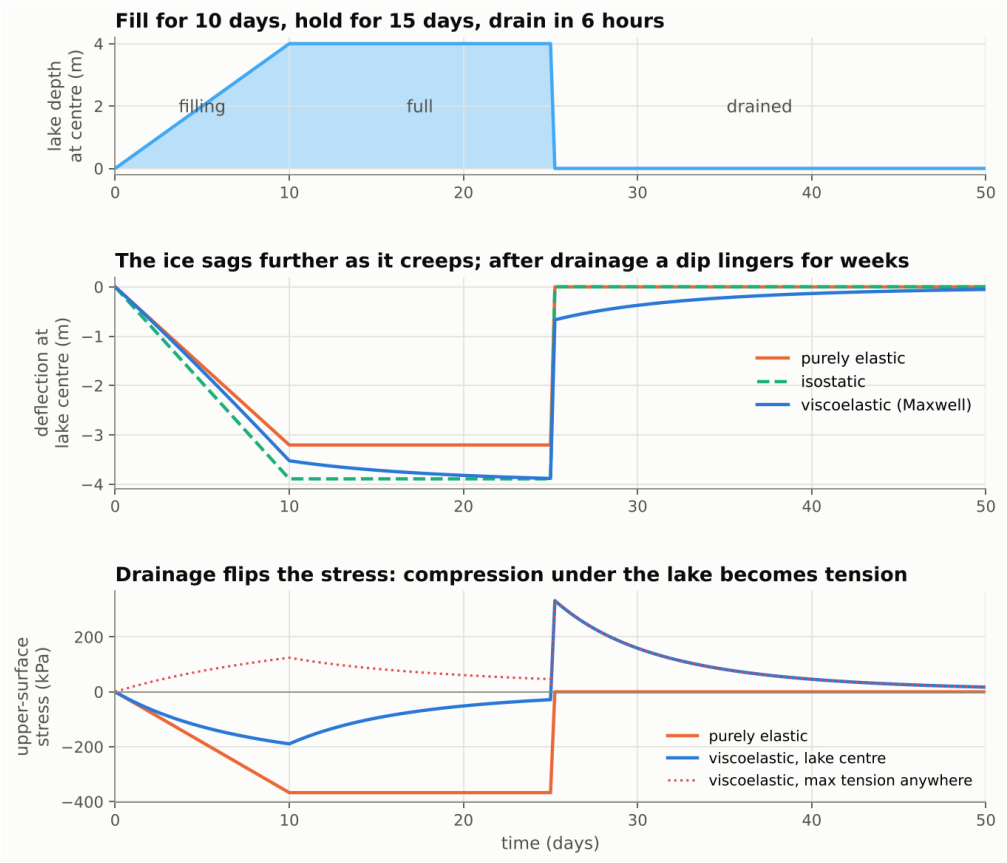


Figure 8: A lake fills for 10 days, is held for 15 days and drains in 6 hours. Top: lake depth. Middle: deflection at the lake centre. Bottom: bending stress at the upper surface.

The viscoelastic twist comes with **drainage**:

- **While the lake sits**, the moments relax ($M \to 0$). The ice "gets used to" being bent.
- **When the lake drains** in hours (for example by hydrofracture), the load vanishes much faster than τ. The response to *removing* the load is again elastic, and equal and opposite to the original elastic response. Superposed on a relaxed state, this leaves the ice **bent upwards**, with the surface under the former lake now in **tension (+330 kPa)**.
- A dip of about 0.7 m lingers at the old lake site and creeps away over the following weeks.

Repeated fill-drain cycles therefore reverse the stress at the lake centre again and again. This fatigue-like loading is what the model was built to explore (compare MacAyeal & Sergienko, 2013; Banwell et al., 2019).

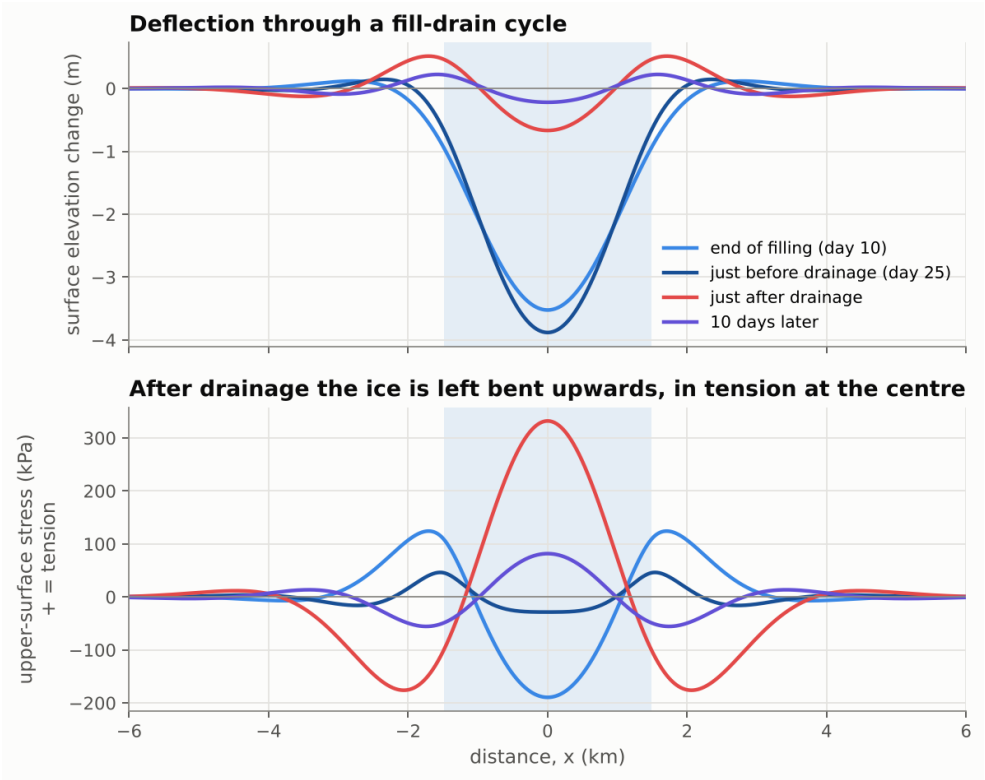


Figure 9: Deflection and upper-surface stress at key moments of the fill-drain cycle.

10 Boundary conditions

The beam equation is fourth order, so each end needs **two** conditions.

end	physical meaning	conditions	type
grounding line	ice frozen to the bed: no deflection, no tilt	$w = 0,\ \partial w/\partial x = 0$ ("clamped")	kinematic: on w
calving front	nothing holds the end	$M = 0,\ V = \partial M/\partial x = 0$ ("free")	dynamic: on M

The free-end conditions are on M, not directly on w. For a Maxwell beam that starts from rest, $M = 0$ for all time implies $\partial^2 w/\partial x^2 = 0$ for all time, and similarly for the third derivative, so the elastic intuition carries over.

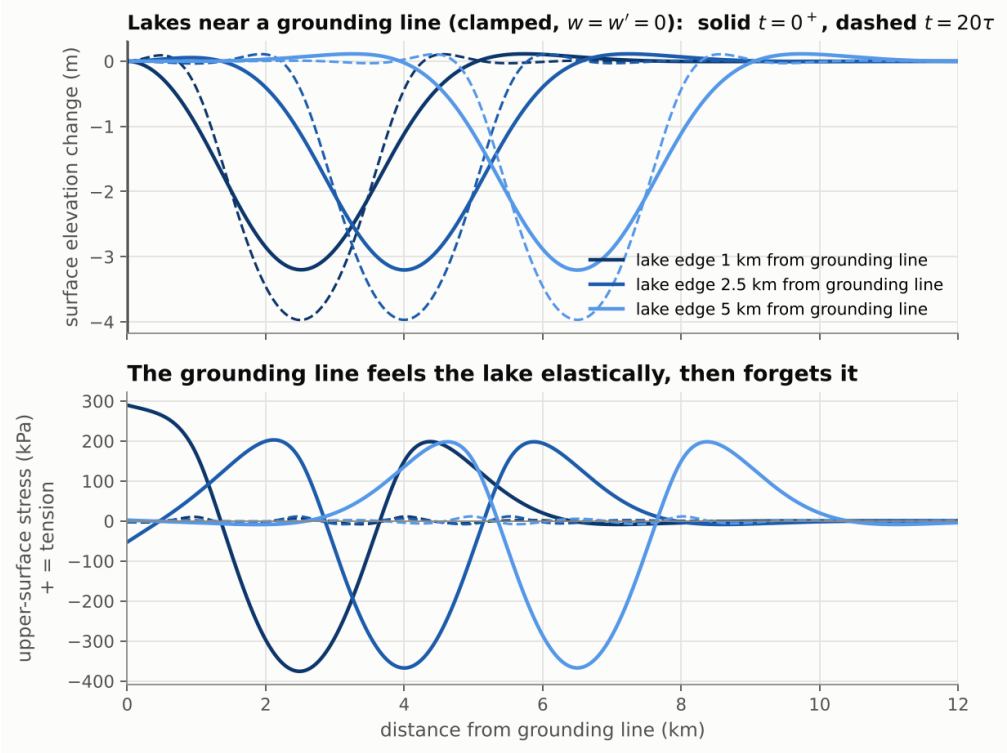


Figure 10: Lakes 1, 2.5 and 5 km from a clamped grounding line. A lake within about 2ℓ of the grounding line strains it elastically: here, +290 kPa of surface tension at the grounding line for the closest lake. That stress relaxes away (dashed curves) as the shelf creeps.

11 Numerics: fast and simple

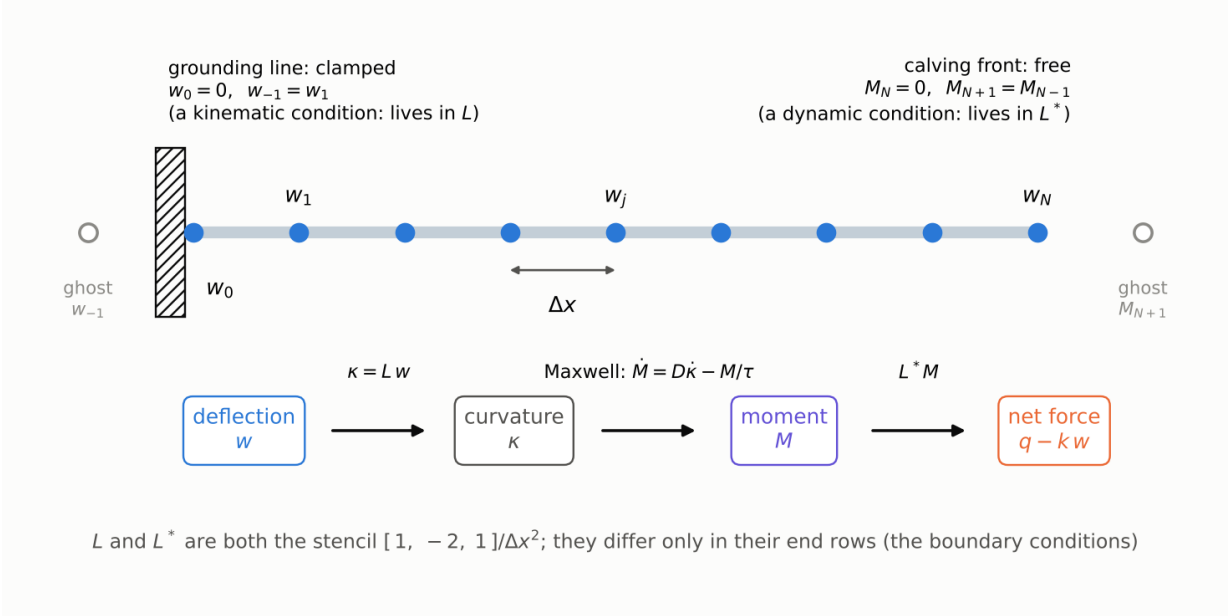


Figure 11: The grid, its ghost nodes and the chain of operators $w \to \kappa \to M \to$ force. Kinematic boundary conditions live in L and dynamic ones in L^*.

11.1 Space: two little matrices

Put N equally spaced nodes x_j along the beam, with spacing Δx, and store w_j and M_j at each node. The physics is a chain of three steps:

$$w \xrightarrow{L} \kappa \xrightarrow{\text{Maxwell}} M \xrightarrow{L^*} \text{net force } (q - kw).$$

Both L and L^* are the familiar second-difference stencil $[1, -2, 1]/\Delta x^2$. They differ **only in their end rows**, which is where the boundary conditions live:

end type	row of L (curvature from w)	row of L^* (force from M)
clamped (node 0)	ghost $w_{-1} = w_1$: $\kappa_0 = (2w_1 - 2w_0)/\Delta x^2$	not used: the equation is replaced by $w_0 = 0$
free (node 0)	not used: zeros, so $M_0 \equiv 0$	ghost $M_{-1} = M_1$: $f_0 = (2M_1 - 2M_0)/\Delta x^2$

Kinematic conditions go in L and dynamic ones go in L^*. The fourth derivative is never written down explicitly. It is just $L^* D\, L$, so a spatially varying $D(x)$ (a thinning shelf) costs nothing extra.

11.2 Time: the trapezoidal rule, then one linear solve

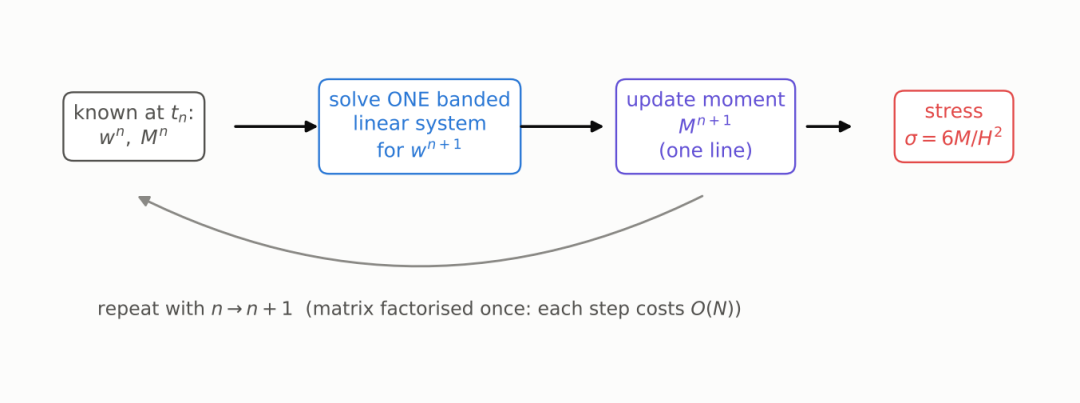


Figure 12: One time step of the algorithm.

Step from t_n to $t_{n+1} = t_n + \Delta t$. For each part $i = 1, 2$ write $a_i = \Delta t/\tau_i$ and $c_i = 1/(1 + a_i/2)$.

1. Moment update. Apply the trapezoidal (Crank–Nicolson) rule to each Maxwell law $\dot{M}_i = D_i L\dot{w} - M_i/\tau_i$ at every node:

$$M_i^{n+1} = c_i \left[\left(1 - \tfrac{a_i}{2}\right) M_i^n + D_i\, L\, (w^{n+1} - w^n) \right].$$

2. Deflection solve. Substitute these into the force balance at the new time, $L^*(M_1^{n+1} + M_2^{n+1}) + k\, w^{n+1} = q^{n+1}$. This gives **one linear system** for the new deflection, with $\bar{D} = c_1 D_1 + c_2 D_2$:

$$\underbrace{\left(\quad L^* \bar{D}\, L + k\, I \quad \right)}_{\text{constant, pentadiagonal}} w^{n+1} = q^{n+1} + L^* \bar{D}\, L\, w^n - L^* \left[\textstyle\sum_i c_i \left(1 - \tfrac{a_i}{2}\right) M_i^n \right].$$

3. Repeat. Solve step 2 for w^{n+1}, then evaluate step 1 for each M_i^{n+1}.

That is the entire algorithm. In the code it reads (`Mp0` holds the two parts, and `c`, `a` have one row per part):

```
def step(self, w0, Mp0, q1, dt):
    lu, S, c, a = self._step_matrices(dt)          # S = L* (c1 D1 + c2 D2) L, factorised on
    rhs = q1 + S @ w0 - self.Lstar @ (c * (1 - a/2) * Mp0).sum(axis=0)
    w1 = self._solve(lu, rhs)                       # one banded back-substitution
    Mp1 = c * ((1 - a/2) * Mp0 + self.Dp * (self.L @ (w1 - w0)))
    return w1, Mp1
```

11.3 Why this is a good scheme

- **Fast.** The matrix never changes, so it is factorised once. Each step is then one banded back-substitution costing $O(N)$. 3000 nodes × 1200 steps take about one second in pure Python and SciPy.
- **Exact elastic limit.** With $\Delta t/\tau \to 0$ it reduces to the elastic solve, so instantaneous loading and unloading are captured exactly.
- **Second-order accurate** in both Δx and Δt, and **unconditionally stable**: the trapezoidal rule is A-stable.
- **Force balance holds to round-off at every step.** The two operators are used consistently, so the discrete version of (1) is satisfied exactly, and the stresses are physically consistent.
- **Choosing steps.** $\Delta x \lesssim \ell/10$ and $\Delta t \lesssim \tau_2/10$ give errors well below 0.1 % (see the convergence panel below).

12 Verification

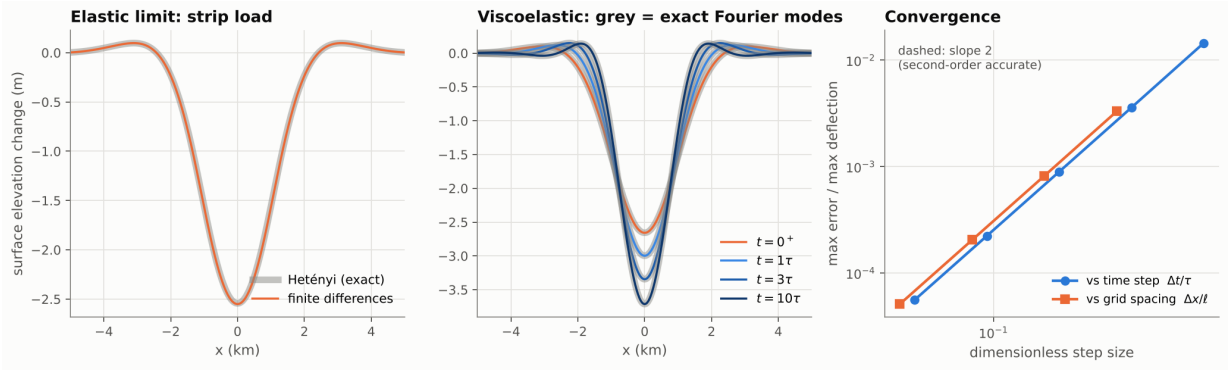


Figure 13: Verification. Left: elastic strip load against Hetényi's exact solution. Centre: viscoelastic relaxation against the exact Fourier-mode solution (grey). Right: second-order convergence in Δt and Δx.

The test-suite (`flexure/tests/test_viscoelastic_beam.py`, run with `python -m pytest -q`) checks the model against exact solutions:

test	exact solution	agreement
elastic strip load, free beam	Hetényi (1946)	< 0.2 %
line load at a free calving front	$w = \frac{2P\beta}{k} e^{-\beta x}\cos\beta x$	< 0.2 %

test	exact solution	agreement
uniform load, clamped grounding line	$w = \frac{q}{k}\left[1 - e^{-\beta x}(\cos\beta x + \sin\beta x)\right]$ (the tidal-flexure profile)	< 0.1 %
viscoelastic relaxation of a lake	exact Fourier-mode solution (§8, two-element)	< 0.2 %
equal Poisson's ratios	the single-element closed form $\tau_\lambda = \tau[1 + (k\ell)^4]$	to round-off
moment tensor in map view	$D[(1-\nu)\kappa + \nu\,\mathrm{tr}\,\kappa\, I]$ elastically, $\Lambda[(1-\nu_v)\dot\kappa + \nu_v\,\mathrm{tr}\,\dot\kappa\, I]$ when viscous	$< 10^{-4}$
long-time limit	local isostasy $w = q/k$, $M \to 0$	✓
uniform load on a free beam	no bending at all	to round-off
discrete force balance $L^*M + kw = q$	(all boundary-condition combinations)	to round-off
time convergence	slope 2	✓

13 Limitations and easy extensions

- **Lake–flexure feedback.** In this document the water depth is prescribed. In reality the lake sits in the depression it creates. The routing notebooks couple the plate to fill-spill-merge, so water and ice shape each other:
  - `flexure/lake_drainage_flexure.ipynb` drains a routed lake;
  - `flexure/coupled_routing_flexure.ipynb` routes a melt season over the flexing shelf;
  - `flexure/drainage_reorganisation.ipynb` follows where the next summer's water goes after a lake drains.
- **Variable thickness.** In 1-D this already works: pass `H` as an array. Both D and Λ scale as H^3, so τ is unchanged. The 2-D plates need a uniform thickness, because their moment potentials assume a uniform rigidity. Carrying the moment tensor instead, as MacAyeal et al. (2015) do, removes that restriction at about the same cost.
- **Lakes in map view.** This is implemented in `flexure/viscoelastic_plate.py`: the same physics with $\partial^4/\partial x^4 \to \nabla^4$ gives a viscoelastic plate. A round lake sinks less than a long trough at first, and the ring of tension survives. See Part 3 of `flexure/flexure_model_tutorial.ipynb`.
- **Nonlinear (Glen's law) creep and depth-varying temperature.** Here η depends on stress and depth, so Λ becomes a function of the state. The moment update becomes a local nonlinear update; the solve stays the same. MacAyeal et al. (2015, eqs 39–40) give a ready-made effective viscosity in terms of the bending moments. It is what keeps their lake basin from recovering after drainage (§5).
- **Membrane stresses and ice-shelf flow.** These are neglected, which is fine for bending over days to weeks. For coupled flow and flexure see MacAyeal et al. (2021).
- **Thin-plate validity for narrow lakes.** The beam results assume $k_x H \ll 1$. At $\lambda = 1$ km on 250 m ice, $k_x H \approx 1.6$. There the beam overestimates τ_λ, and the $(2\pi\ell/\lambda)^4$ scaling becomes unreliable (§8.5).
- **Fracture.** The model reports stresses but does not break.

14 Parameters and references

symbol	meaning	default
H	ice thickness	250 m
E	effective Young's modulus (tidal-flexure studies infer about 1 GPa, well below the laboratory 9 GPa; e.g. Vaughan, 1995)	1 GPa
ν	elastic Poisson's ratio	0.3
ν_v	Poisson's ratio of viscous creep (incompressible)	0.5
η	ice viscosity	10^{14} Pa s
ρ_i, ρ_{sw}, ρ_w	ice, sea water, meltwater densities	917, 1028, 1000 kg m^{-3}
$D = EH^3/12(1-\nu^2)$	flexural rigidity	1.4×10^{15} N m
$k = \rho_{sw} g$	buoyancy stiffness	1.0×10^4 Pa m^{-1}
$\ell = (D/k)^{1/4}$	flexural length	614 m
$\tau = 4\eta(1-\nu^2)/E$	mean Maxwell time	4.2 days
$\tau_1 = 4\eta(1+\nu_v)(1-\nu)/E$	relaxation time, isotropic part	4.9 days
$\tau_2 = 4\eta(1-\nu_v)(1+\nu)/E$	relaxation time, deviatoric part	3.0 days

The values of E and especially η are uncertain by an order of magnitude or more. Treat the numbers above as illustrative. Because $\ell \propto E^{1/4} H^{3/4}$ and $\tau \propto \eta/E$, it is easy to rescale the results.

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